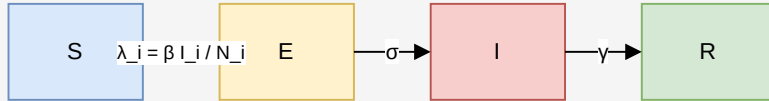


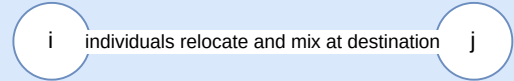
Metapopulation reaction-diffusion dynamics

Within-city dynamics (SEIR shown)

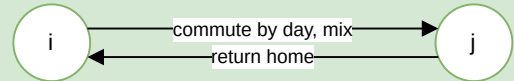


variants: $SIS (I \rightarrow S)$, $SQIR (I \rightarrow Q \rightarrow R)$, $SEIRS (R \rightarrow S)$, $SEIQR+D$

Diffusive mobility (air, water)



Recurrent mobility (land)



couples force of infection:
 $\pi_i = \sum_j C_{ij} \cdot I^*_j / N^*_j$

Each day: local reaction, then network mobility

With no land layer, $C = I$ and $\pi_i = I_i / N_i$ (plain diffusive coupling).